7 SEPTEMBER 2026 - 28 FEBRUARY 2027

Junior Automation and Controls Technician Syllabus

A focused technical curriculum with modules, activities, exercises, case studies and competency gates

DESIGN PROMISE

Technical learning only. Portfolio construction, CV work and applications are intentionally kept outside this syllabus.

- 14 technical modules
- 25 core weeks
- Approximately 425 guided hours
- PC-first practical work with explicit safety limits

Prepared for Emmanual Januarie

Study period: 7 September 2026 to 31 March 2027

Workload: 17 hours per full week

Purpose and target standard

This syllabus prepares Emmanual Januarie to contribute as a junior or trainee automation and control technician under supervision. It emphasizes electrical controls, PLCs, instrumentation, HMI/SCADA, industrial communications, systematic troubleshooting, testing and technical documentation.

HONEST BOUNDARY

This is a personal training syllabus, not a trade qualification, apprenticeship, licence, site authorization, safety certificate or substitute for supervised plant experience.

Role-aligned competency priorities

Priority	What a junior should be able to demonstrate
Core	Read basic control schematics; trace 24 VDC I/O; explain the PLC scan; create and test Ladder Logic; work with modes, permissives, interlocks and sequences.
Core	Diagnose faults across power, field device, wiring, I/O, logic, output, actuator, process, network and HMI layers without random changes.
Core	Create usable I/O lists, tag registers, control narratives, cause-and-effect tables, network maps, test records, fault reports and change logs.
Role-dependent	Scale 4-20 mA signals, interpret measurement faults, understand valves and PID behavior, and support HMI/SCADA, trends and industrial communications.
Awareness	Know when a task requires formal authorization, qualified electrical work, site procedures, OEM training, cybersecurity controls or escalation.

The role scan behind this design consistently emphasizes PLC/HMI/SCADA, drives, instrumentation, industrial networks, drawings, testing, commissioning, fault finding, documentation, safety and supervised learning. The syllabus therefore gives these areas more weight than advanced theory or vendor-specific specialization.

Program map

No.	Dates	Technical module	Primary output
1	7-13 Sep 2026	Electrical fundamentals and safe work boundaries	Calculation sheet; One-page fault report
2	14-20 Sep 2026	Industrial components, control circuits and schematics	Motor-control schematic; Troubleshooting report
3	21 Sep-4 Oct 2026	PLC hardware, scan cycle, addressing and I/O	Architecture drawing; Backup record
4	5-11 Oct 2026	Ladder Logic core patterns	Commented routine library; Revision log
5	12 Oct-1 Nov 2026	Modes, permissives, interlocks, sequences and faults	Sequence chart; FAT results
6	2-15 Nov 2026	Motors, starters, protection and VFD fundamentals	Starter schematic; Test record
7	16-29 Nov 2026	Sensors and measurement principles	Process sketch; Trend-based fault report
8	30 Nov-13 Dec 2026	Analog signals, 4-20 mA, scaling and calibration records	Loop diagram; Three-point verification
9	14 Dec 2026-3 Jan 2027	Process control, valves, alarms and PID fundamentals	P&ID; Cause-and-effect row
10	4-10 Jan 2027	HMI and SCADA design, alarms and trends	Screen hierarchy; Comms-loss screenshots
11	11-24 Jan 2027	Industrial networking and control communications	Network architecture; Fault report
12	25-31 Jan 2027	SCADA integration, diagnostics and historical data	Architecture; Recovery record
13	1-7 Feb 2027	Maintenance troubleshooting, testing and change control	Ten fault reports; Handover note
14	8-28 Feb 2027	System integration, commissioning and lifecycle handover	Integration specification; Handover

MODULE COMPLETION RULE

A module is complete only when the learner can explain the concept, apply it, test normal and fault behavior, and produce readable evidence. Watching material alone is preparation, not completion.

Assessment and evidence standard

Each module uses the same evidence ladder so claims remain defensible.

EVIDENCE LABELS	COMPLETION GATE
[ ] Studied - explain in your own words	[ ] Closed-book explanation completed
[ ] Simulated - built in training software	[ ] Exercises corrected and understood
[ ] Implemented - configured/programmed yourself	[ ] Normal behavior tested
[ ] Tested - expected and actual results recorded	[ ] At least two fault paths tested
[ ] Documented - usable technical record exists	[ ] Root cause supported by evidence
[ ] Certified - valid formal certificate held	[ ] Files dated, versioned and backed up

Technician troubleshooting method

Stage	Required behavior
1. Make safe	Identify hazards, stored energy, state and authorization limits.
2. Define	Write expected behavior, actual behavior, timing and recent changes.
3. Check basics	Power, safety state, mode, permissives, device status, links and alarms.
4. Trace	Field device -> I/O -> tag -> logic -> output -> actuator -> process -> HMI.
5. Isolate	Use evidence and known-good comparison; change one thing at a time.
6. Test	Write the expected safe result before performing the test.
7. Diagnose	Name the root cause separately from symptoms and contributing conditions.
8. Correct	Make one authorized, controlled correction with rollback where relevant.
9. Verify	Repeat the failed step, normal test, fault test and recovery test.
10. Document	Record cause, action, result, version, remaining risk and handover.

MODULE	DATES	DURATION
01 - Electrical fundamentals and safe work boundaries	7-13 Sep 2026	1 week

Build the electrical vocabulary, calculations, schematic-reading habits, and safety boundaries needed before PLC or field work.

ROLE VALUE

Essential foundation for every automation, PLC, SCADA and instrumentation role.

CORE CONTENT

[ ] Voltage, current, resistance, power, energy, AC/DC, polarity and frequency

[ ] Series and parallel circuits; open, short, overload and earth-fault concepts

[ ] Electrical symbols, current-path tracing, test points and multimeter functions

[ ] Isolation, LOTO awareness, test-before-touch and authorization limits

REQUIRED ACTIVITIES

[ ] Build a quantities map with symbol, unit, instrument and meaning

[ ] Draw and calculate one series and one parallel 24 VDC circuit

[ ] Predict then simulate normal, open and short-circuit behavior

[ ] Write a safe-measurement checklist for an unknown low-voltage circuit

Required exercises

[ ] Calculate current and power for 24 VDC across 240 ohms

[ ] Explain why voltage can be present with no current

[ ] Mark five safe test points on a control circuit

Module outcome

By the end of this module, the learner must explain the core content without copied notes, complete the exercises, apply the topic in simulation or a safe training context, diagnose the case study systematically, and produce the listed deliverables.

SAFETY AND SCOPE CHECK

State assumptions and authorization limits. Never use mains voltage, defeat protection, force live equipment, or represent simulation as commissioned field work.

Module 01 technician case study

A control-panel indicator is dark

Situation: Supply output is 24 VDC. Voltage before the fuse is 24 VDC and after it is 0 VDC. The indicator resistance is plausible.

Available evidence: Power-supply reading, before/after-fuse readings, component resistance

Your tasks

[ ] Draw the expected current path

[ ] Identify the most likely failed layer

[ ] Name the next safe confirmation test

[ ] State correction and recovery verification

Required technical evidence

DELIVERABLES	VERIFICATION RECORD
[ ] Calculation sheet [ ] Annotated circuit [ ] Expected-vs-actual table [ ] One-page fault report	[ ] Expected result written before testing [ ] Normal condition tested [ ] At least two relevant faults tested [ ] Recovery tested [ ] Actual results recorded [ ] Untested items marked NOT TESTED [ ] Files versioned and backed up [ ] Result explainable without notes

CASE-STUDY RULE

Do not guess or change multiple things at once. Define expected behavior, isolate the failing layer, choose a safe test, support the root cause with evidence, correct it, verify recovery and record the result.

[ ] MODULE COMPLETE Date: ____________________ Revisit needed: YES / NO

MODULE	DATES	DURATION
02 - Industrial components, control circuits and schematics	14-20 Sep 2026	1 week

Recognize common control-panel components and trace command, protection and feedback paths through a basic motor-control circuit.

ROLE VALUE

Directly supports panel reading, motor controls and PLC I/O troubleshooting.

CORE CONTENT

[ ] NO/NC states, relays, contactors, auxiliary contacts and terminal blocks

[ ] Fuses, breakers, overloads, control versus power circuit

[ ] Two-wire and three-wire control; seal-in circuits

[ ] Fail-safe stops, interface relays, E-stop and safety-relay awareness

REQUIRED ACTIVITIES

[ ] Label a DOL start/stop schematic

[ ] Build a state table for stopped, starting, running and overload

[ ] Convert relay behavior into PLC input/logic/output terms

[ ] Color-code command, protection and feedback paths

Required exercises

[ ] Explain why Stop is commonly NC

[ ] Trace Start press to latched run

[ ] List three causes of motor-will-not-start

Module outcome

By the end of this module, the learner must explain the core content without copied notes, complete the exercises, apply the topic in simulation or a safe training context, diagnose the case study systematically, and produce the listed deliverables.

SAFETY AND SCOPE CHECK

State assumptions and authorization limits. Never use mains voltage, defeat protection, force live equipment, or represent simulation as commissioned field work.

Module 02 technician case study

Contactor drops out when Start is released

Situation: The coil energizes while Start is held. Stop and overload are healthy. The holding contact never changes state.

Available evidence: Coil state, auxiliary-contact state, stop and overload states

Your tasks

[ ] Trace the seal-in path

[ ] List two plausible root causes

[ ] Design one test that distinguishes them

[ ] Record verified recovery

Required technical evidence

DELIVERABLES	VERIFICATION RECORD
[ ] Motor-control schematic	[ ] Expected result written before testing
[ ] Component list	[ ] Normal condition tested
[ ] State table	[ ] At least two relevant faults tested
[ ] Troubleshooting report	[ ] Recovery tested
	[ ] Actual results recorded
	[ ] Untested items marked NOT TESTED
	[ ] Files versioned and backed up
	[ ] Result explainable without notes

CASE-STUDY RULE

Do not guess or change multiple things at once. Define expected behavior, isolate the failing layer, choose a safe test, support the root cause with evidence, correct it, verify recovery and record the result.

[ ] MODULE COMPLETE Date: ____________________ Revisit needed: YES / NO

MODULE	DATES	DURATION
03 - PLC hardware, scan cycle, addressing and I/O	21 Sep-4 Oct 2026	2 weeks

Understand where signals enter and leave a PLC, how the scan executes, and how to trace a signal end to end.

ROLE VALUE

Core PLC and control-systems technician competency.

CORE CONTENT	REQUIRED ACTIVITIES
[ ] CPU, power, local/remote I/O and communications modules	[ ] Draw a PLC-field-HMI architecture
[ ] Input scan, program solve, output update and housekeeping	[ ] Create a 12-point I/O list and tag standard
[ ] Digital and analog I/O; physical addresses, tags and internal memory	[ ] Trace one input and one output through every layer
[ ] PNP/NPN awareness, module diagnostics, monitoring, forcing risk and backups	[ ] Simulate a short pulse and relate it to scan time

Required exercises

[ ] Classify Start, contactor, transmitter and valve command by I/O type

[ ] Explain the scan cycle in five sentences

[ ] List causes for sensor-on but PLC-input-off

Module outcome

By the end of this module, the learner must explain the core content without copied notes, complete the exercises, apply the topic in simulation or a safe training context, diagnose the case study systematically, and produce the listed deliverables.

SAFETY AND SCOPE CHECK

State assumptions and authorization limits. Never use mains voltage, defeat protection, force live equipment, or represent simulation as commissioned field work.

Module 03 technician case study

Sensor LED is on but the PLC input is off

Situation: The sensor is powered and detects the object. The input module LED is off, the PLC tag is false, and other channels work.

Available evidence: Sensor LED, module LED, tag state, healthy neighboring channels

Your tasks

[ ] Separate device, wiring, module, tag and HMI possibilities

[ ] Order checks from safest to most intrusive

[ ] Select the likely boundary

[ ] Define the pass/fail recovery test

Required technical evidence

DELIVERABLES	VERIFICATION RECORD
[ ] Architecture drawing	[ ] Expected result written before testing
[ ] I/O list	[ ] Normal condition tested
[ ] Tag standard	[ ] At least two relevant faults tested
[ ] Signal-trace report	[ ] Recovery tested
[ ] Backup record	[ ] Actual results recorded
	[ ] Untested items marked NOT TESTED
	[ ] Files versioned and backed up
	[ ] Result explainable without notes

CASE-STUDY RULE

Do not guess or change multiple things at once. Define expected behavior, isolate the failing layer, choose a safe test, support the root cause with evidence, correct it, verify recovery and record the result.

[ ] MODULE COMPLETE Date: ____________________ Revisit needed: YES / NO

MODULE	DATES	DURATION
04 - Ladder Logic core patterns	5-11 Oct 2026	1 week

Read, write, comment and test maintainable Ladder Logic before building operating modes and sequences.

ROLE VALUE

Demonstrates the minimum programming fluency expected in junior PLC roles.

CORE CONTENT

[ ] Rung evaluation, examine-on/off, coils and internal bits

[ ] Seal-in, Set/Reset, one-shots and reset risks

[ ] On-delay/off-delay timers, counters and comparisons

[ ] Retentive behavior, duplicate coils, comments and written tests

REQUIRED ACTIVITIES

[ ] Write a start/stop rung from memory

[ ] Build a five-second delayed output

[ ] Build an object counter with one-shot and reset

[ ] Write truth tables before running each routine

Required exercises

[ ] Compare seal-in with Set/Reset

[ ] Predict a timer when enable flickers

[ ] Find and correct a duplicate-coil fault

Module outcome

By the end of this module, the learner must explain the core content without copied notes, complete the exercises, apply the topic in simulation or a safe training context, diagnose the case study systematically, and produce the listed deliverables.

SAFETY AND SCOPE CHECK

State assumptions and authorization limits. Never use mains voltage, defeat protection, force live equipment, or represent simulation as commissioned field work.

Module 04 technician case study

One box is counted several times

Situation: The photoeye remains true for 180 ms, the scan is about 10 ms, and the counter is enabled directly by the input.

Available evidence: Input duration, scan estimate, counter enable logic

Your tasks

[ ] Explain the scan-related cause

[ ] Choose an edge pattern

[ ] Predict the corrected count

[ ] Add a sensor-chatter test

Required technical evidence

DELIVERABLES	VERIFICATION RECORD
[ ] Commented routine library [ ] Truth tables [ ] Timer/counter tests [ ] Revision log	[ ] Expected result written before testing [ ] Normal condition tested [ ] At least two relevant faults tested [ ] Recovery tested [ ] Actual results recorded [ ] Untested items marked NOT TESTED [ ] Files versioned and backed up [ ] Result explainable without notes

CASE-STUDY RULE

Do not guess or change multiple things at once. Define expected behavior, isolate the failing layer, choose a safe test, support the root cause with evidence, correct it, verify recovery and record the result.

[ ] MODULE COMPLETE Date: ____________________ Revisit needed: YES / NO

MODULE	DATES	DURATION
05 - Modes, permissives, interlocks, sequences and faults	12 Oct-1 Nov 2026	3 weeks

Design predictable command ownership, ordered steps, fault handling and controlled recovery.

ROLE VALUE

A high-value competence across packaging, process and machine automation.

CORE CONTENT	REQUIRED ACTIVITIES
[ ] Command, status, feedback, permissive, interlock, alarm and trip	[ ] Create a permissive summary and Boolean start condition
[ ] Auto/Manual and Local/Remote ownership; transition behavior	[ ] Define ownership in every mode
[ ] Sequence steps, transitions and state-machine awareness	[ ] Build a three-step sequence
[ ] Start-failure timing, first-out fault, acknowledgement, reset and cause-and-effect	[ ] Inject five faults and prove reset requires correction

Required exercises

[ ] Classify low pressure as permissive, alarm or trip

[ ] Design Auto-to-Manual transition behavior

[ ] Explain acknowledgement versus reset

Module outcome

By the end of this module, the learner must explain the core content without copied notes, complete the exercises, apply the topic in simulation or a safe training context, diagnose the case study systematically, and produce the listed deliverables.

SAFETY AND SCOPE CHECK

State assumptions and authorization limits. Never use mains voltage, defeat protection, force live equipment, or represent simulation as commissioned field work.

Module 05 technician case study

A conveyor stalls at transfer

Situation: Step 3 is active. Safety and overload are healthy. Downstream Ready is false. No trip is active.

Available evidence: Active step, safety state, overload state, readiness input

Your tasks

[ ] Explain why code changes are not the first action

[ ] Trace Downstream Ready

[ ] Define useful HMI diagnostics

[ ] Write recovery and regression tests

Required technical evidence

DELIVERABLES	VERIFICATION RECORD
[ ] Sequence chart	[ ] Expected result written before testing
[ ] Control narrative	[ ] Normal condition tested
[ ] Cause-and-effect	[ ] At least two relevant faults tested
[ ] PLC logic	[ ] Recovery tested
[ ] FAT results	[ ] Actual results recorded
	[ ] Untested items marked NOT TESTED
	[ ] Files versioned and backed up
	[ ] Result explainable without notes

CASE-STUDY RULE

Do not guess or change multiple things at once. Define expected behavior, isolate the failing layer, choose a safe test, support the root cause with evidence, correct it, verify recovery and record the result.

[ ] MODULE COMPLETE Date: ____________________ Revisit needed: YES / NO

MODULE	DATES	DURATION
06 - Motors, starters, protection and VFD fundamentals	2-15 Nov 2026	2 weeks

Understand motor-control interfaces and distinguish PLC, starter, drive, motor and process faults safely.

ROLE VALUE

Matches common technician duties around motors, drives and production downtime.

CORE CONTENT

[ ] Three-phase motor and nameplate fundamentals

[ ] DOL/reversing starters, interlocks and overload feedback

[ ] VFD run, direction, speed, ready, running and fault signals

[ ] Hardwired, analog and network interfaces; ramps and stored-energy awareness

REQUIRED ACTIVITIES

[ ] Trace a PLC-to-DOL command and feedback loop

[ ] Read and explain a sample motor nameplate

[ ] Create a PLC-to-VFD signal map

[ ] Build a motor faceplate and layer-based fault tree

Required exercises

[ ] Explain command versus running feedback

[ ] Calculate percent speed at 35 Hz of 50 Hz

[ ] List reasons a healthy drive may not run

Module outcome

By the end of this module, the learner must explain the core content without copied notes, complete the exercises, apply the topic in simulation or a safe training context, diagnose the case study systematically, and produce the listed deliverables.

SAFETY AND SCOPE CHECK

State assumptions and authorization limits. Never use mains voltage, defeat protection, force live equipment, or represent simulation as commissioned field work.

Module 06 technician case study

VFD has a run command but the motor does not turn

Situation: PLC command is true and speed reference is 60 percent. The drive shows LOCAL and Ready is false.

Available evidence: PLC command, speed reference, control source and Ready status

Your tasks

[ ] Identify the controlling layer

[ ] Explain why PLC edits are inappropriate

[ ] State the simulated correction

[ ] Verify start, stop, speed and fault response

Required technical evidence

DELIVERABLES	VERIFICATION RECORD
[ ] Starter schematic	[ ] Expected result written before testing
[ ] VFD signal map	[ ] Normal condition tested
[ ] PLC routine	[ ] At least two relevant faults tested
[ ] Faceplate	[ ] Recovery tested
[ ] Test record	[ ] Actual results recorded
	[ ] Untested items marked NOT TESTED
	[ ] Files versioned and backed up
	[ ] Result explainable without notes

CASE-STUDY RULE

Do not guess or change multiple things at once. Define expected behavior, isolate the failing layer, choose a safe test, support the root cause with evidence, correct it, verify recovery and record the result.

[ ] MODULE COMPLETE Date: ____________________ Revisit needed: YES / NO

MODULE	DATES	DURATION
07 - Sensors and measurement principles	16-29 Nov 2026	2 weeks

Understand how physical conditions become trustworthy values and how selection, installation and process effects influence measurement.

ROLE VALUE

Strengthens instrumentation awareness without claiming calibration authorization.

CORE CONTENT	REQUIRED ACTIVITIES
[ ] Process-sensor-transmitter-PLC-HMI chain	[ ] Draw four measurement chains
[ ] Temperature, pressure, level and flow technologies	[ ] Compare three technologies for one service
[ ] LRV, URV, range, span, accuracy, repeatability and response	[ ] Create an instrument list
[ ] Open, short, frozen, noisy, drifting and implausible failures	[ ] Classify six trends by likely failure symptom

Required exercises

[ ] Select level technology for clean water and foaming service

[ ] Calculate span for -1 to 9 bar

[ ] Explain accurate instrument but poor measurement

Module outcome

By the end of this module, the learner must explain the core content without copied notes, complete the exercises, apply the topic in simulation or a safe training context, diagnose the case study systematically, and produce the listed deliverables.

SAFETY AND SCOPE CHECK

State assumptions and authorization limits. Never use mains voltage, defeat protection, force live equipment, or represent simulation as commissioned field work.

Module 07 technician case study

Ultrasonic level jumps during filling

Situation: The level is erratic only while the inlet valve is open. Communication is healthy; foam and turbulence form below the sensor.

Available evidence: Trend behavior, communication quality, observed foam/turbulence

Your tasks

[ ] Separate electrical, configuration, installation and process causes

[ ] Choose the likely cause

[ ] Propose mitigations

[ ] Define evidence of improvement

Required technical evidence

DELIVERABLES	VERIFICATION RECORD
[ ] Process sketch	[ ] Expected result written before testing
[ ] Instrument list	[ ] Normal condition tested
[ ] Selection matrix	[ ] At least two relevant faults tested
[ ] Trend-based fault report	[ ] Recovery tested
	[ ] Actual results recorded
	[ ] Untested items marked NOT TESTED
	[ ] Files versioned and backed up
	[ ] Result explainable without notes

CASE-STUDY RULE

Do not guess or change multiple things at once. Define expected behavior, isolate the failing layer, choose a safe test, support the root cause with evidence, correct it, verify recovery and record the result.

[ ] MODULE COMPLETE Date: ____________________ Revisit needed: YES / NO

MODULE	DATES	DURATION
08 - Analog signals, 4-20 mA, scaling and calibration records	30 Nov-13 Dec 2026	2 weeks

Convert raw signals into engineering units, recognize abnormal values and troubleshoot analog loops systematically.

ROLE VALUE

A core junior instrumentation and PLC integration skill.

CORE CONTENT	REQUIRED ACTIVITIES
[ ] 4-20 mA live zero and transmitter wiring concepts	[ ] Create percent/current tables for two ranges
[ ] Raw counts, current, percent and engineering-unit scaling	[ ] Build a two-way scaling worksheet
[ ] Under/over-range, open loop, noise, shielding and filtering	[ ] Simulate two transmitters and compare values
[ ] Zero/span, as-found/as-left records and HART awareness	[ ] Add abnormal-signal and plausibility alarms

Required exercises

[ ] Convert 12 mA on a 0-10 bar range

[ ] Convert 650 L/min to current for 0-1000 L/min

[ ] Explain 0 mA versus valid 0 percent

Module outcome

By the end of this module, the learner must explain the core content without copied notes, complete the exercises, apply the topic in simulation or a safe training context, diagnose the case study systematically, and produce the listed deliverables.

SAFETY AND SCOPE CHECK

State assumptions and authorization limits. Never use mains voltage, defeat protection, force live equipment, or represent simulation as commissioned field work.

Module 08 technician case study

Pressure display is consistently high

Situation: A 0-10 bar transmitter gives about 12 mA at trusted 5 bar, while PLC scaling is configured for 0-16 bar.

Available evidence: Transmitter range, current, trusted reference and PLC scaling

Your tasks

[ ] Calculate the correct value

[ ] Identify the root cause

[ ] Explain why recalibration is wrong

[ ] Verify 0, 50 and 100 percent

Required technical evidence

DELIVERABLES	VERIFICATION RECORD
[ ] Loop diagram	[ ] Expected result written before testing
[ ] Scaling workbook	[ ] Normal condition tested
[ ] PLC configuration	[ ] At least two relevant faults tested
[ ] Alarm tests	[ ] Recovery tested
[ ] Three-point verification	[ ] Actual results recorded
	[ ] Untested items marked NOT TESTED
	[ ] Files versioned and backed up
	[ ] Result explainable without notes

CASE-STUDY RULE

Do not guess or change multiple things at once. Define expected behavior, isolate the failing layer, choose a safe test, support the root cause with evidence, correct it, verify recovery and record the result.

[ ] MODULE COMPLETE Date: ____________________ Revisit needed: YES / NO

MODULE	DATES	DURATION
09 - Process control, valves, alarms and PID fundamentals	14 Dec 2026-3 Jan 2027	3 weeks

Understand feedback-loop behavior so process symptoms, actuator faults and alarm responses can be interpreted correctly.

ROLE VALUE

Provides process literacy needed to troubleshoot beyond the PLC code.

CORE CONTENT

[ ] Process objective, PV, SP, MV/CV and disturbances

[ ] Open/closed loop, Manual/Automatic and bumpless-transfer awareness

[ ] Proportional, integral, derivative, lag, dead time, overshoot and saturation

[ ] Valve action/fail position; alarm deadband, priority, trip and recovery

REQUIRED ACTIVITIES

[ ] Draw three feedback loops

[ ] Interpret stable, overshoot, oscillation and saturation trends

[ ] Create an alarm list with operator response

[ ] Test Manual/Automatic level behavior and valve discrepancy

Required exercises

[ ] Identify PV, SP and MV in a level loop

[ ] Compare High alarm and High-High trip

[ ] Write controlled restart after a trip

Module outcome

By the end of this module, the learner must explain the core content without copied notes, complete the exercises, apply the topic in simulation or a safe training context, diagnose the case study systematically, and produce the listed deliverables.

SAFETY AND SCOPE CHECK

State assumptions and authorization limits. Never use mains voltage, defeat protection, force live equipment, or represent simulation as commissioned field work.

Module 09 technician case study

Tank reaches High-High while outlet is commanded open

Situation: Outlet command is 100 percent, position feedback remains 5 percent, inlet flow continues, and communications are healthy.

Available evidence: Command, position feedback, inlet flow, level trend and comms state

Your tasks

[ ] Identify the failed loop element

[ ] State the protective action

[ ] Choose diagnostic trends

[ ] Write recovery verification

Required technical evidence

DELIVERABLES	VERIFICATION RECORD
[ ] P&ID	[ ] Expected result written before testing
[ ] Control narrative	[ ] Normal condition tested
[ ] Alarm list	[ ] At least two relevant faults tested
[ ] Trend analysis	[ ] Recovery tested
[ ] Cause-and-effect row	[ ] Actual results recorded
	[ ] Untested items marked NOT TESTED
	[ ] Files versioned and backed up
	[ ] Result explainable without notes

CASE-STUDY RULE

Do not guess or change multiple things at once. Define expected behavior, isolate the failing layer, choose a safe test, support the root cause with evidence, correct it, verify recovery and record the result.

[ ] MODULE COMPLETE Date: ____________________ Revisit needed: YES / NO

MODULE	DATES	DURATION
10 - HMI and SCADA design, alarms and trends	4-10 Jan 2027	1 week

Present process state clearly and make mode, interlock, quality and communications problems visible to operators.

ROLE VALUE

Directly supports junior SCADA and controls support work.

CORE CONTENT	REQUIRED ACTIVITIES
[ ] Overview, detail, alarm, trend and diagnostic displays	[ ] Sketch a five-screen hierarchy
[ ] Equipment faceplates, ownership, navigation and restrained color	[ ] Build a process overview and equipment faceplate
[ ] Alarm priority, acknowledgement and operator response	[ ] Configure five alarms with responses
[ ] Tag quality, timestamps, stale data, access and command confirmation	[ ] Create a diagnostic trend with command, feedback, PV, SP and quality

Required exercises

[ ] Choose overview versus maintenance information

[ ] Specify behavior during comms loss

[ ] Design confirmation for a high-impact command

Module outcome

By the end of this module, the learner must explain the core content without copied notes, complete the exercises, apply the topic in simulation or a safe training context, diagnose the case study systematically, and produce the listed deliverables.

SAFETY AND SCOPE CHECK

State assumptions and authorization limits. Never use mains voltage, defeat protection, force live equipment, or represent simulation as commissioned field work.

Module 10 technician case study

A normal-looking value remains after PLC communications fail

Situation: The network is removed. Level remains at 62 percent; timestamp stops; comms status is false but hidden.

Available evidence: Last value, frozen timestamp, hidden comms status

Your tasks

[ ] Identify the display failure

[ ] Define stale-data appearance

[ ] Add status and alarm behavior

[ ] Test loss and recovery

Required technical evidence

DELIVERABLES	VERIFICATION RECORD
[ ] Screen hierarchy [ ] Overview [ ] Faceplate [ ] Alarm tests [ ] Comms-loss screenshots	[ ] Expected result written before testing [ ] Normal condition tested [ ] At least two relevant faults tested [ ] Recovery tested [ ] Actual results recorded [ ] Untested items marked NOT TESTED [ ] Files versioned and backed up [ ] Result explainable without notes

CASE-STUDY RULE

Do not guess or change multiple things at once. Define expected behavior, isolate the failing layer, choose a safe test, support the root cause with evidence, correct it, verify recovery and record the result.

[ ] MODULE COMPLETE Date: ____________________ Revisit needed: YES / NO

MODULE	DATES	DURATION
11 - Industrial networking and control communications	11-24 Jan 2027	2 weeks

Diagnose physical, link, addressing, protocol, mapping and application problems in the correct order.

ROLE VALUE

Industrial networks are repeatedly requested in current control-system roles.

CORE CONTENT

[ ] Ethernet links, switches, topology, MAC, IPv4, subnet and gateway

[ ] Ping, TCP/UDP and client/server awareness

[ ] RS-485, Modbus RTU/TCP, registers, offsets, data types and byte order

[ ] OPC UA concepts; Profinet/EtherNet-IP awareness; timeouts and stale data

REQUIRED ACTIVITIES

[ ] Draw PLC-HMI-SCADA-switch-VFD architecture

[ ] Create an IP schedule and subnet checks

[ ] Build a ten-point Modbus map

[ ] Run timeout, bad-map and recovery tests

Required exercises

[ ] Check whether two /24 addresses share a subnet

[ ] Explain IP address versus register address

[ ] List causes when ping works but data does not

Module outcome

By the end of this module, the learner must explain the core content without copied notes, complete the exercises, apply the topic in simulation or a safe training context, diagnose the case study systematically, and produce the listed deliverables.

SAFETY AND SCOPE CHECK

State assumptions and authorization limits. Never use mains voltage, defeat protection, force live equipment, or represent simulation as commissioned field work.

Module 11 technician case study

SCADA can ping the PLC but tank level does not update

Situation: Pump status updates. PLC level is 57.3 percent. The client reads 40011 while the map assigns level to 40010.

Available evidence: Successful ping, working pump status, PLC value and conflicting register map

Your tasks

[ ] Name the working and failing layers

[ ] Correct the mapping

[ ] Verify type and scale

[ ] Test timeout and recovery

Required technical evidence

DELIVERABLES	VERIFICATION RECORD
[ ] Network architecture	[ ] Expected result written before testing
[ ] IP schedule	[ ] Normal condition tested
[ ] Protocol map	[ ] At least two relevant faults tested
[ ] Comms test record	[ ] Recovery tested
[ ] Fault report	[ ] Actual results recorded
	[ ] Untested items marked NOT TESTED
	[ ] Files versioned and backed up
	[ ] Result explainable without notes

CASE-STUDY RULE

Do not guess or change multiple things at once. Define expected behavior, isolate the failing layer, choose a safe test, support the root cause with evidence, correct it, verify recovery and record the result.

[ ] MODULE COMPLETE Date: ____________________ Revisit needed: YES / NO

MODULE	DATES	DURATION
12 - SCADA integration, diagnostics and historical data	25-31 Jan 2027	1 week

Prove the complete process-to-PLC-to-network-to-SCADA path, including commands, alarms, trends, quality and recovery.

ROLE VALUE

Builds credible end-to-end SCADA support capability.

CORE CONTENT

[ ] End-to-end tag lifecycle and naming

[ ] Read/write permissions, confirmation and audit awareness

[ ] Alarm/event path; scan, polling, trend and historian concepts

[ ] Watchdogs, backup/export/restore and least-privilege awareness

REQUIRED ACTIVITIES

[ ] Integrate at least 20 tags

[ ] Create a tag register with source, scale, access and quality

[ ] Test five alarms end to end

[ ] Break communication, restore it and record recovery

Required exercises

[ ] Trace one operator command and feedback

[ ] Specify read-only and read/write tags

[ ] Write communications acceptance criteria

Module outcome

By the end of this module, the learner must explain the core content without copied notes, complete the exercises, apply the topic in simulation or a safe training context, diagnose the case study systematically, and produce the listed deliverables.

SAFETY AND SCOPE CHECK

State assumptions and authorization limits. Never use mains voltage, defeat protection, force live equipment, or represent simulation as commissioned field work.

Module 12 technician case study

Values update but operator commands are rejected

Situation: Status values update. Start has no effect. The Start tag is configured read-only; local physical start works.

Available evidence: Healthy connection, read-only tag configuration, working local command

Your tasks

[ ] Separate network, access, PLC and field layers

[ ] Correct the tag access

[ ] Add confirmation and audit behavior

[ ] Test allowed and denied commands

Required technical evidence

DELIVERABLES	VERIFICATION RECORD
[ ] Architecture	[ ] Expected result written before testing
[ ] Tag register	[ ] Normal condition tested
[ ] Alarm tests	[ ] At least two relevant faults tested
[ ] Trends	[ ] Recovery tested
[ ] Command-path tests	[ ] Actual results recorded
[ ] Recovery record	[ ] Untested items marked NOT TESTED
	[ ] Files versioned and backed up
	[ ] Result explainable without notes

CASE-STUDY RULE

Do not guess or change multiple things at once. Define expected behavior, isolate the failing layer, choose a safe test, support the root cause with evidence, correct it, verify recovery and record the result.

[ ] MODULE COMPLETE Date: ____________________ Revisit needed: YES / NO

MODULE	DATES	DURATION
13 - Maintenance troubleshooting, testing and change control	1-7 Feb 2027	1 week

Diagnose with evidence, preserve recoverability and document controlled changes and handovers.

ROLE VALUE

Troubleshooting discipline is the strongest transferable technician signal.

CORE CONTENT	REQUIRED ACTIVITIES
[ ] Symptom, root cause, contributing condition and known-good state	[ ] Build an expected-state table
[ ] Layered trace: power-field-I/O-logic-output-actuator-process-HMI	[ ] Create a check-basics-first fault flow
[ ] Monitoring, trends, alarms, forcing/bypass risks	[ ] Inject and diagnose ten faults
[ ] Backup, version, rollback, FAT/SAT, retest and escalation	[ ] Write a controlled-change and unresolved-handover template

Required exercises

[ ] Separate symptom and cause in five examples

[ ] Explain how forcing hides faults

[ ] Write normal, fault and recovery tests

Module outcome

By the end of this module, the learner must explain the core content without copied notes, complete the exercises, apply the topic in simulation or a safe training context, diagnose the case study systematically, and produce the listed deliverables.

SAFETY AND SCOPE CHECK

State assumptions and authorization limits. Never use mains voltage, defeat protection, force live equipment, or represent simulation as commissioned field work.

Module 13 technician case study

Pump will not start after maintenance

Situation: Auto is selected but Start Permitted is false. E-stop, overload and level are healthy. Local/Remote feedback says Local.

Available evidence: Permissive summary, selector feedback, healthy PLC/network

Your tasks

[ ] Define expected state

[ ] Identify the blocking permissive

[ ] Explain why bypass is wrong

[ ] Restore, verify and hand over

Required technical evidence

DELIVERABLES	VERIFICATION RECORD
[ ] Ten fault reports	[ ] Expected result written before testing
[ ] Expected-state tables	[ ] Normal condition tested
[ ] Change log	[ ] At least two relevant faults tested
[ ] Backup record	[ ] Recovery tested
[ ] Handover note	[ ] Actual results recorded
	[ ] Untested items marked NOT TESTED
	[ ] Files versioned and backed up
	[ ] Result explainable without notes

CASE-STUDY RULE

Do not guess or change multiple things at once. Define expected behavior, isolate the failing layer, choose a safe test, support the root cause with evidence, correct it, verify recovery and record the result.

[ ] MODULE COMPLETE Date: ____________________ Revisit needed: YES / NO

MODULE	DATES	DURATION
14 - System integration, commissioning and lifecycle handover	8-28 Feb 2027	3 weeks

Combine technical subsystems into a traceable, testable control solution without turning the syllabus into a portfolio module.

ROLE VALUE

Completes the technical lifecycle expected of a supervised junior contributor.

CORE CONTENT

[ ] Requirements, scope, assumptions, exclusions and acceptance criteria

[ ] Equipment modules, modes, commands, status, faults and safe states

[ ] Interface control: tags, units, scaling, ownership and network-loss behavior

[ ] FAT/SAT awareness, defect control, backups, cybersecurity hygiene and handover

REQUIRED ACTIVITIES

[ ] Freeze a small integration specification

[ ] Audit one tag across drawings, PLC, HMI and SCADA

[ ] Execute normal, mode, fault, comms-loss and recovery tests

[ ] Prepare a concise technical handover and limitation statement

Required exercises

[ ] Write five measurable acceptance criteria

[ ] Decide what remains local if SCADA fails

[ ] Build a defect-and-retest workflow

Module outcome

By the end of this module, the learner must explain the core content without copied notes, complete the exercises, apply the topic in simulation or a safe training context, diagnose the case study systematically, and produce the listed deliverables.

SAFETY AND SCOPE CHECK

State assumptions and authorization limits. Never use mains voltage, defeat protection, force live equipment, or represent simulation as commissioned field work.

Module 14 technician case study

SCADA fails during an automatic fill

Situation: PLC scan and local I/O remain healthy. High and High-High protection live in the PLC. Remote commands are unavailable and displayed values become stale.

Available evidence: PLC health, local protection, stale SCADA values and lost remote command

Your tasks

[ ] Define what continues locally

[ ] Specify indication and ownership

[ ] Choose pause/continue/stop behavior from risk

[ ] Test protection and recovery

Required technical evidence

DELIVERABLES	VERIFICATION RECORD
[ ] Integration specification	[ ] Expected result written before testing
[ ] Interface register	[ ] Normal condition tested
[ ] FAT	[ ] At least two relevant faults tested
[ ] Defect register	[ ] Recovery tested
[ ] Backup/restore record	[ ] Actual results recorded
[ ] Handover	[ ] Untested items marked NOT TESTED
	[ ] Files versioned and backed up
	[ ] Result explainable without notes

CASE-STUDY RULE

Do not guess or change multiple things at once. Define expected behavior, isolate the failing layer, choose a safe test, support the root cause with evidence, correct it, verify recovery and record the result.

[ ] MODULE COMPLETE Date: ____________________ Revisit needed: YES / NO

Final technical release standard

The target is credible junior contribution under supervision, not unsupported expert status.

TARGET LEVEL 3/4	TARGET LEVEL 2/3
[ ] Electrical calculations and schematic tracing	[ ] HMI/SCADA, alarms, trends and quality
[ ] PLC scan, I/O and Ladder Logic	[ ] Industrial networks and protocol diagnostics
[ ] Modes, permissives, sequences and faults	[ ] Motors, VFDs, valves and PID awareness
[ ] 4-20 mA scaling and loop reasoning	[ ] Lifecycle documents, FAT and change control
[ ] Systematic troubleshooting and verification	[ ] Cybersecurity and authorization awareness

Release questions

[ ] Can every screenshot and diagram be explained without notes?

[ ] Does each test show expected and actual results?

[ ] Are simulations labelled as simulations?

[ ] Are unperformed tests marked NOT TESTED?

[ ] Can one complete signal be traced through field, PLC, network and HMI layers?

[ ] Can a fault be isolated without random code changes?

[ ] Are safety and authorization limits stated clearly?

Role-alignment references

Content was cross-checked against the attached source PDFs and a current role scan on 2 September 2026. Job requirements vary by employer, industry and jurisdiction; each vacancy and local legal framework remains authoritative.

Current role examples

1. Adsyst Automation - Graduate/Junior Control Systems Technician: PLC, SCADA, HMI, VSDs, instrumentation, specifications, architecture, testing and commissioning.

https://careers.adsyst.co.uk/job/886184

2. Siemens - Field Service Engineer Apprenticeship: electrical safety, drawings, troubleshooting, PLC/HMI operation, networks, FAT/SAT and document control.

https://jobs.siemens.com/en_US/externaljobs/JobDetail/478759

3. Anglo American / De Beers - C&I; Technician Plant (South Africa): instrumentation and control maintenance, PLC/SCADA programming and problem-solving methods; formal qualification and experience requirements apply.

https://jobs.smartrecruiters.com/AngloAmericanDeBeersGroup/744000143313275-c-i-technician-plant

IMPORTANT

This study plan improves technical readiness and evidence quality. It does not guarantee employment or replace employer-specific qualifications, trade requirements, site inductions, medicals, licences or supervised experience.